

Constellation Lens

User Manual

A Network Analysis Engine for Knowledge Discovery

Cognitive Engine — Layer 3

Version 1.0 — April 2026

uConstellation.world

What Is the Constellation Lens?

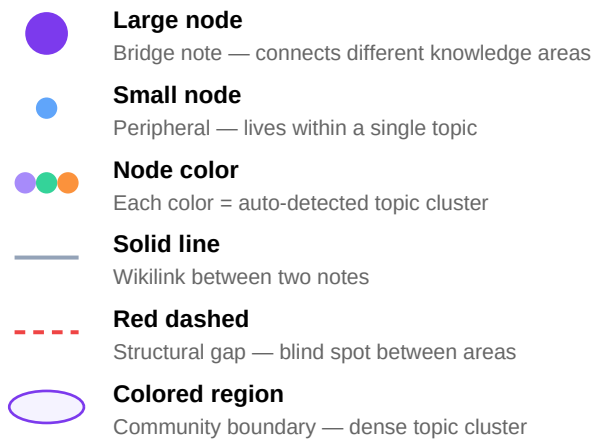
Imagine looking at a city from above at night. Some buildings have many roads connecting them — they are hubs. Some neighborhoods are dense with activity, while others sit isolated. And between some neighborhoods, there are no roads at all — blind spots that could benefit from a bridge.

The Constellation Lens does this for your knowledge. It takes your notes and the links between them and analyzes their structure using algorithms from network science. It answers: **"What patterns and gaps exist in my thinking?"**

It does not tell you what to think. It shows you the shape of what you already know — where your knowledge is deep, where it is shallow, and where two areas could connect but do not yet.

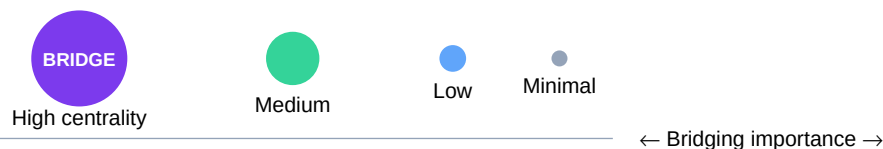
Visual Legend

Every element in the Lens graph has meaning:



Node Size = Bridging Importance

The Lens resizes every node based on its **betweenness centrality** — a measure of how many shortest paths between other notes pass through it. A note with only 3 links can be a critical bridge if it connects two large topic clusters that would otherwise be disconnected.



Node Color = Topic Community

The Lens uses the **Louvain algorithm** to automatically detect topic clusters (communities) from your link patterns — not from folders or tags. Each community gets a distinct color. Notes of the same color are densely connected to each other.



Structural Gaps = Blind Spots

The most distinctive insight: the Lens identifies pairs of communities with **high internal density but low inter-community connections**. These are blind spots — areas where two relevant knowledge domains exist but lack bridges between them. Based on Ronald Burt's structural holes theory (1992).



Universe Health

A composite score (0-100) measuring the cognitive diversity of your knowledge base:



0.72
MODULARITY

10%
DOMINANCE

5.50
ENTROPY

2.7
LINKS/NOTE

Component	What It Measures	Healthy Range
Modularity	How distinct your topic clusters are	0.3 – 0.6
Dominance	% of notes in the largest community	Below 35%
Entropy	How evenly knowledge is distributed	Above 2.0 bits
Links/Note	Average connections per note	Above 1.0

How to Use

1. Click the **Lens button** in the left dock bar (magnifier with plus sign)
2. Wait a few seconds while the analysis runs
3. The Lens view appears: graph on the left, analytics panel on the right
4. Explore: check the health score, review the top bridges, examine communities
5. Look for blind spots — each is a research opportunity
6. Click any bridge note to open it in the editor
7. Close: click × or press Escape

Tips

Start with the health score. If it is low, check dominance and entropy to understand why.

Check the blind spots. Each gap between communities is a potential research question.

Watch the bridges. Your top bridge notes are the structural backbone — develop them further.

Use layer peeling. Hide dominant MOC notes to reveal the organic structure underneath.

The Lens works better with more links. The more wikilinks you create, the richer the analysis.

Advanced Features

Layer Peeling: Temporarily hide the top 1-20 most central notes. This reveals the secondary structure beneath dominant MOC and index notes.

Tag Edges: Toggle on to reveal implicit connections between notes sharing tags but no wikilinks. Often reveals connections you have been making unconsciously.

How It Works

Betweenness Centrality (Brandes' algorithm, 2001): For each note, counts how many shortest paths between all other note pairs pass through it. Computed in Rust. $O(VE)$ complexity.

Community Detection (Louvain algorithm): Groups notes into clusters by maximizing modularity. Computed in JavaScript.

Structural Gap Detection (Burt's structural holes, 1992): Identifies community pairs with high internal density but low inter-community connections.

All computation runs **locally on your machine**. No data leaves your device.